

A reversing game in Japanese picture books: the role of prosodic constituents

Language games have provided insights into the phonological structure of various languages (e.g. Bagemihl 1995). For example, one language game widely played by Japanese children is *shiritori*, in which a player is expected to come up with a word that begins with the last “unit” of the word uttered by the previous player (e.g. [go.ri.**ra**] → [**rap**.pa]). The unit used in *shiritori* is a mora (Katada 1990), and hence this game highlights the prominent role of moras in Japanese prosodic organization (e.g. Labrune 2012). However, there is also a report that pre-literate children can use syllables rather than moras in *shiritori* (e.g. [ne.ku.**tai**] → [**tai**.ja.ki]: Kubozono 2000), suggesting the possibility that the prominence of moras in Japanese adult phonology may be related to the mastery of Japanese orthographic knowledge, which is mora-based (Inagaki et al. 2000).

The primary aim of the present paper is to describe and analyze a hitherto undocumented language game pattern found in the *henshin-series* picture books authored by Tadashi Akiyama. This pattern resembles that of *zuu-ja-go* pattern, a reversing argot played by Japanese musicians (Ito et al. 1996), but it shows some interesting differences. The language game involves reversing of sounds in existing words to transform them into a different word. One example is shown in Figure 1, in which a frog making [ge.ko-ge.ko] sounds becomes [ko.ge], a burnt frog. Akiyama published more than 15 books of this series. By analyzing the examples in these books, we found that syllables play a crucial role in many examples, although other prosodic units can occasionally be relevant.



Figure 1: An example taken from *Henshin Otenki* “Transformational weather”. Akiyama, Tadashi (2016). Kin-no-hoshi-sha.

The simplest pattern is when the original words consist of two light syllables, in which case the two light syllables are reversed, as in (1). (Throughout this abstract, syllable boundaries are shown with a dot “.”) For these examples, we cannot determine whether the unit of reversal is syllables or moras, because the boundaries of these two prosodic units coincide.

- (1) [go.ma] → [ma.go] [mi.na] → [na.mi]
 [ro.bo] → [bo.ro] [wa.ku] → [ku.wa]
 [ne.ba] → [ba.ne] [bu.ka] → [ka.bu]

We found a number of examples in which the original words consist of three light syllables. In some cases, the initial syllables are moved to the end, as in (2a), whereas in other cases, the final syllables are placed word-initially, as in (2b).

- (2a) [na.o.to] → [o.to.na]
 [na.o.ki] → [o.ki.na]
 [ja.su.si] → [su.si.ja]
- (2b) [ka.zu.o] → [o.ka.zu]
 [ta.tsu.ko] → [ko.ta.tsu]
 [sa.me.ko] → [ko.sa.me]

One interesting generalization that emerged is that when the original word contains a heavy syllable, that syllable is almost always the target of reversal, as illustrated in (3)-(6).

- (3) CV_iV_i (long vowel)
- | | |
|---|---|
| [ru. bii] → [bii .ru] | [zuu .go] → [go. zuu] |
| [su.to. roo] → [roo .su.to] | [zuu .na.na] → [na.na. zuu] |

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- [juu.zi] → [zi.juu] [oo.ka.mi] → [ka.mi.oo]
- (4) CVN (moraic nasal)
 [ga.man] → [man.ga] [ken.zi] → [zi.ken]
 [ka.ku.san] → [san.ka.ku] [tom.bo] → [bo.ton]
 [bo.tan] → [tam.bo] [kan.na] → [na.kan]
- (5) CVQ (geminate)
 [pa.ka?] → [kap.pa] [kap.pu] → [pu.ka?]
 [ka.ma?] → [mak.ka] [rap.pa] → [pa.ra?]
- (6) CV_iV_j (diphthong)
 [ma.kou] → [kou.ma] [tai.ja.ki] → [ja.ki.tai]
 [to.kei] → [kei.to]
 [ja.tai] → [tai.ja]

When the words consist of two heavy syllables, these two syllables are reversed, as in (7). We found one example in which a light syllable is followed by two heavy syllables; in this case, the last heavy syllable is moved to the word-initial position ([go.roo.zin] → [zin.go.roo]).

- (7) [zii.san] → [san.zii] [san.sai] → [sai.san] [mon.dee] → [dee.mon]
 [man.zuu] → [zuu.man] [tai.hou] → [hou.tai] [kin.zoo] → [zoo.kin]

What we do not find is a case in which one sub-constituent mora of a heavy syllable is moved to achieve the reversal in words like (3)-(7). For syllables containing a moraic nasal [N] and a geminate [Q], lack of such patterns may be explained in terms of Japanese phonotactics, as neither [N] or [Q] can appear word-initially. There are no words that begin with these sounds in the first place, and moreover, [N] and [Q] in word-final positions cannot be moved to word-initial positions. For syllables containing long vowels and diphthongs, however, moving a portion of these sequences would not violate Japanese phonotactics; e.g. [oo.ka.mi] could have been changed into [o.ka.mi.o], a pattern that is not attested. As such, it seems safe to conclude that this language game primarily operates at the level of syllables. Syllables are treated as indivisible chunks in the patterns in (3)-(7). We find this generalization to be particularly interesting, because of the prominent role of moras in other Japanese language games, such as *shiritori*.

An alternative analysis using a bimoraic foot (Poser 1990) may be possible, except that for forms like (2), we would have to assume that words like [na(oto)] and [na(oki)] have a foot that is aligned with the right edge of the word. This sort of foot structure is unmotivated insofar as it does not align with the morphological boundary in these words (i.e. [nao+to] and [nao+ki]). To complete the picture, while the major unit of reversal seems to be the syllable, when the original words consist of one heavy syllable, mora-based reversal can occur, as in (8).

- (8) [wai] → [i.wa] [kaa] → [a.ka]
 [waa] → [a.wa] [kai] → [i.ka]

Taken together, it may be the case that the game operates at different prosodic levels, with syllables playing the major role—regardless, one clear conclusion is that it operates on prosodic constituents: what moves is either a mora, a syllable or a foot. The current analysis of this hitherto undescribed language game pattern implies that the author Akiyama may be, whether consciously or not, intending to promote the phonological awareness of Japanese children through his books.